C++ CONDITIONAL STATEMENTS

Assignment

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Section: D

Programming Fundamentals

Task 1: Check Whether a Number is Positive or Negative

C++ Source Code:

```
#include <iostream>
using namespace std;

int main()
{
    int num;

    cout << "Enter a number: ";
    cin >> num;

    if (num >= 0)
        cout << "Positive";
    else
        cout << "Negative";

    return 0;
}
```

Sample Output:

```
Sample Input: 8
Sample Output: Positive
```

Task 2: Check Whether a Number is Positive, Negative, or Zero

C++ Source Code:

```
#include <iostream>
using namespace std;

int main()
{
    int num;

    cout << "Enter a number: ";
    cin >> num;

    if (num > 0)
        cout << "Positive";
    else if (num < 0)
        cout << "Negative";
    else
        cout << "Zero";

    return 0;
}
```

Sample Output:

```
Sample Input: 0
Sample Output: Zero
```

Task 3: Calculate Grade from Marks

C++ Source Code:

```
#include <iostream>
using namespace std;

int main()
{
    int marks;

    cout << "Enter marks: ";
    cin >> marks;

    if (marks >= 80)
        cout << "Grade A";
    else if (marks >= 70)
        cout << "Grade B";
    else if (marks >= 60)
        cout << "Grade C";
    else
        cout << "Grade D";

    return 0;
}
```

Sample Output:

```
Sample Input: 75
Sample Output: Grade B
```

Task 4A: Find the Largest of Three Numbers Using Nested if-else

C++ Source Code:

```
#include <iostream>
using namespace std;

int main()
{
    int a, b, c;

    cout << "Enter three numbers: ";
    cin >> a >> b >> c;

    if (a > b)
    {
        if (a > c)
            cout << "Largest = " << a;
        else
            cout << "Largest = " << c;
    }
    else
    {
        if (b > c)
            cout << "Largest = " << b;
        else
            cout << "Largest = " << c;
    }

    return 0;
}
```

Sample Output:

```
Sample Input: 12 25 18
Sample Output: Largest = 25
```

Task 4B: Check Positive Number and Count Its Digits

C++ Source Code:

```
#include <iostream>
using namespace std;

int main()
{
    int num, count = 0;

    cout << "Enter a number: ";
    cin >> num;

    if (num > 0)
    {
        int temp = num;

        while (temp > 0)
        {
            count++;
            temp = temp / 10;
        }

        cout << "Positive" << endl;
        cout << "Number of digits = " << count;
    }
    else
    {
        cout << "Number is not positive";
    }

    return 0;
}
```

Sample Output:

```
Sample Input: 4567
Sample Output:
Positive
Number of digits = 4
```